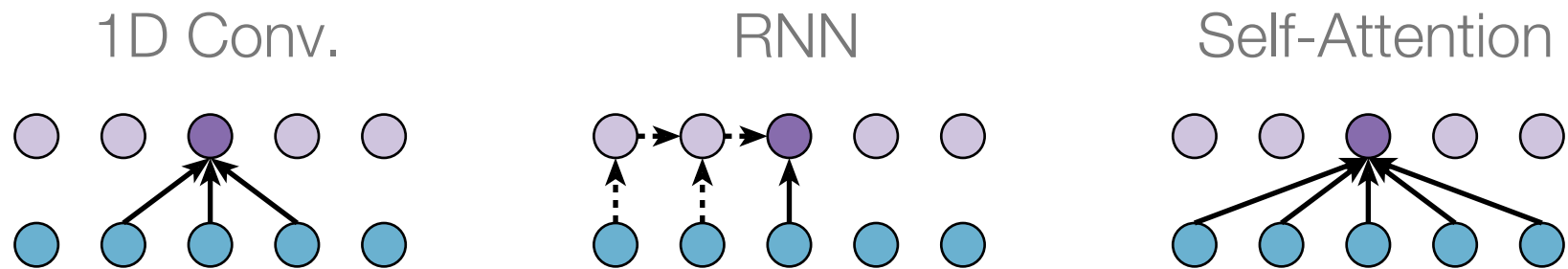


Deep Learning for Time Series

Session 6: State Space Models

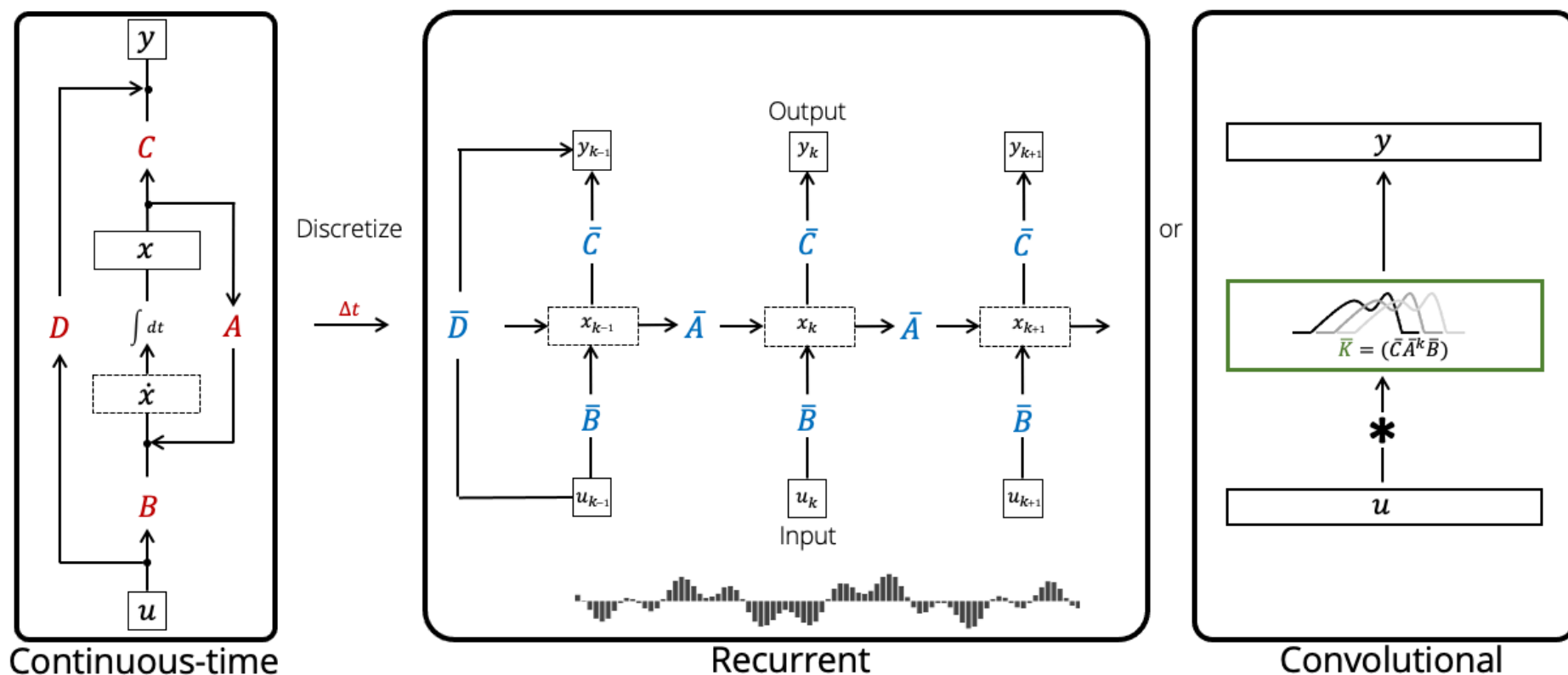
Romain Tavenard

Motivation



- RNNs: sequential, limits parallelism
- CNNs: local receptive field
- Attention-based models: $O(T^2)$ complexity

- Continuous-time dynamics
- Long-range dependencies
- Linear (or near-linear) complexity



Source: “Structured State Spaces: Combining Continuous-Time, Recurrent, and Convolutional Models”, a blogpost by by Albert Gu et al. (2022)

State Space Models

Latent state $h_t \in \mathbb{R}^d$ evolves over time following:

$$\begin{cases} \dot{h}(t) = Ah(t) + Bx(t) \\ o(t) = Ch(t) + \cancel{Dx(t)} \end{cases}$$

- $x(t)$: input
- $o(t)$: output
- $h(t)$: latent state

In practice, we often set $D = 0$ for simplicity (can be later recovered through residual connection in Deep SSM anyway).

SSMs

$$\dot{h}(t) = Ah(t) + Bx(t)$$

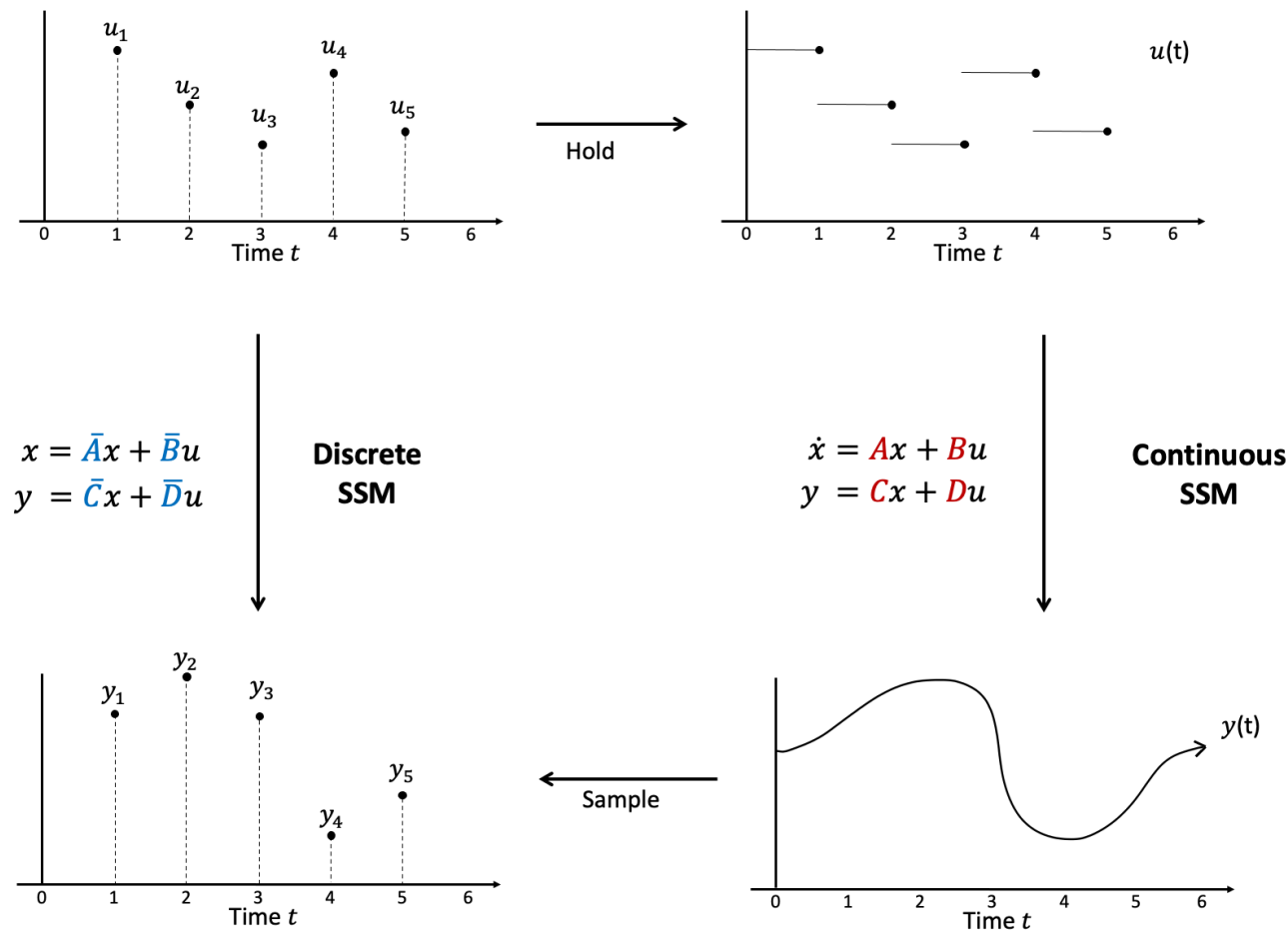
- Linear dynamics specified with respect to the hidden state $h(t)$

Neural ODEs

$$\dot{x}(t) = f_{\theta}(x(t))$$

- Non-linear dynamics specified with respect to the input $x(t)$

Step 1: Time discretization (S4)



Source: "Structured State Spaces: Combining Continuous-Time, Recurrent, and Convolutional Models", a blogpost by by Albert Gu et al. (2022)

$$\dot{h}(t) = Ah(t) + Bx(t)$$

Goal: discretize time with step Δ between indexes $t-1$ and t .

1. Textbook ODE solving gives the following form for $h(t)$:

$$h(t) = e^{A\Delta}h(t - \Delta) + \int_0^{\Delta} e^{As}Bx(t - s) \, ds$$

$$\dot{h}(t) = Ah(t) + Bx(t)$$

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1. Textbook ODE solving gives the following form for $h(t)$:

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2. Assume $x(t)$ is constant between $t - \Delta$ and t :

$$h(t) = e^{A\Delta}h(t - \Delta) + \left(\int_0^{\Delta} e^{As} \, ds \right) Bx(t)$$

$$\dot{h}(t) = Ah(t) + Bx(t)$$

Goal: discretize time with step Δ between indexes $t-1$ and t .

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$$h(t) = e^{A\Delta}h(t - \Delta) + \left(\int_0^{\Delta} e^{As} \, ds \right) Bx(t)$$

3. Get the discrete update rule:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t$$

Now we have the discrete system:

$$\begin{cases} h_t = \bar{A}h_{t-1} + \bar{B}x_t \\ o_t = \bar{C}h_t \end{cases}$$

with:

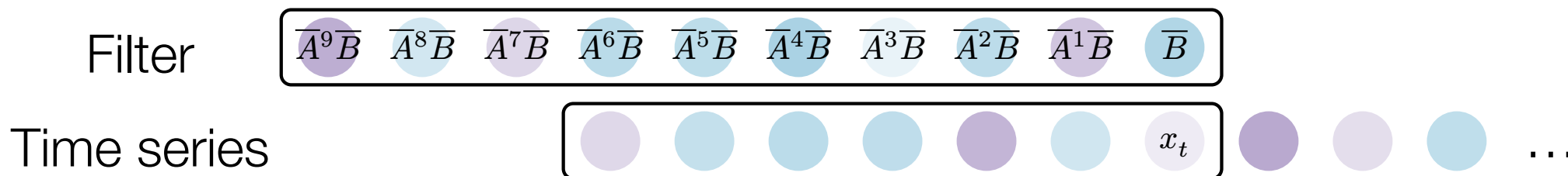
- $\bar{A} = e^{A\Delta}$
- $\bar{B} = \left(\int_0^\Delta e^{As} ds \right) B$
- $\bar{C} = C$

Main idea behind S4: Parametrize A , B , C such that the resulting discrete system has good properties (efficient computation, stable training, long memory).

Efficient computation: The convolutional trick

Let us assume $h_0 = 0$, then we get:

$$h_t = \sum_{k=1}^t \overline{A}^{t-k} \overline{B} x_k$$



Efficient implementation:

- Easy-to-compute powers of $\overline{A}$ (cf. parametrization, next slide)
- Compute the convolution using FFT

Long-term memory: The parametrization trick

- Risk of vanishing for long-range terms (i.e., $\overline{A}^k \overline{B}$ could vanish for large k)
- S4's solution: smart parametrization of A
 - Typical choice: $A = V \Lambda V^{-1}$ with eigenvalues of the form

$$\lambda_j(A) = - \underbrace{\alpha_j}_{>0} + i\omega_j$$

- $\overline{A} = e^{A\Delta} = V e^{\Lambda\Delta} V^{-1}$ with $\lambda_j(\overline{A}) = e^{-\alpha_j\Delta} e^{i\omega_j\Delta}$
- $\overline{A}^k = V (e^{\Lambda\Delta})^k V^{-1}$
- Small $\alpha_j \Rightarrow e^{-\alpha_j\Delta} \approx 1 \Rightarrow$ long-term memory

Step 2: Input-dependent dynamics (Mamba)

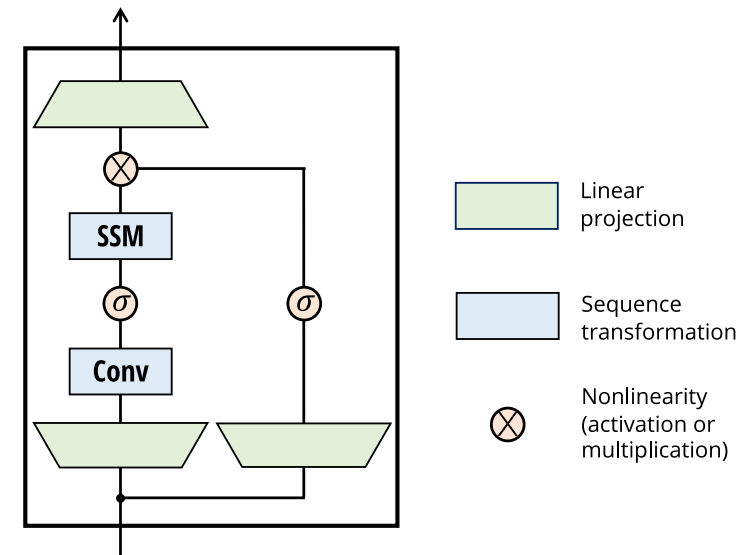
Pure linearity in SSMs:

- ✓ Great long-range memory
- ✗ Limited expressivity

Solution: Make dynamics input-dependent

$$\begin{cases} h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t \\ o_t = \bar{C}_t h_t \end{cases}$$

Now the state transition adapts to the input as in attention-based models



Source: “Mamba: Linear-Time Sequence Modeling with Selective State Spaces”, COLM’24

Step 2: Input-dependent dynamics (Mamba)

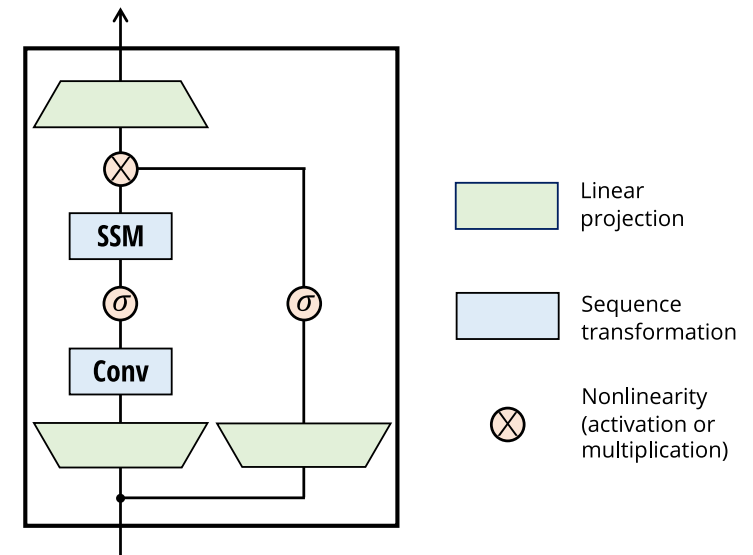
$$\begin{cases} h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t \\ o_t = \bar{C}_t h_t \end{cases}$$

where:

$$\bar{A}_t = \exp(\Delta_t A)$$

$$\bar{B}_t = (\exp(\Delta_t A) - I)(\Delta_t A)^{-1} \Delta_t B$$

and $\Delta_t, \bar{C}_t$ are functions of x_t .



Source: “Mamba: Linear-Time Sequence Modeling with Selective State Spaces”, COLM’24

The challenge: input-dependent parameters break the convolutional structure.

Mamba's design choices:

1. Scan-based computation

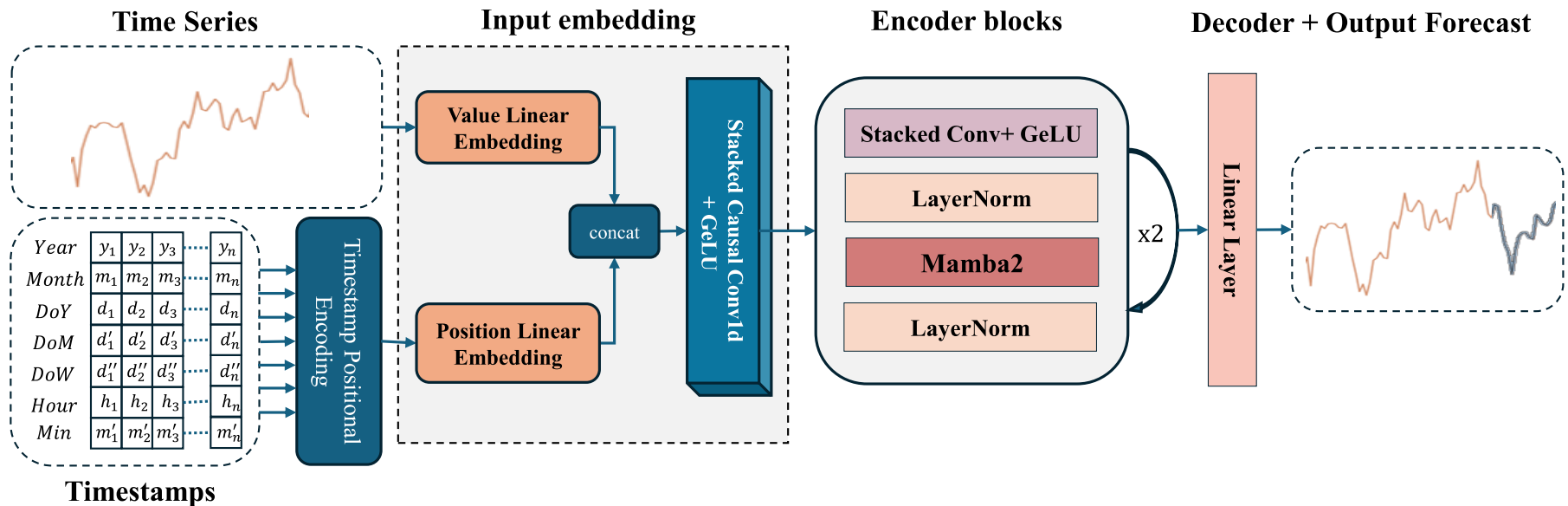
- Process sequentially (like RNN)
- Still efficient thanks to low-level optimizations (e.g., GPU kernels) using the linearity of the state update

2. Gating mechanisms

- Controls what information is retained in the state
- Similar to LSTM/GRU gates

⇒ $O(T)$ complexity

Mamba4Cast: a foundation model for State Space Models (univariate) TS forecasting



Source: “Mamba4Cast: Efficient Zero-Shot Time Series Forecasting with State Space Models”, NeurIPS’24 Workshop